

Lagrangian Variable Cosmology final-v1.2: LVC as an Organising Principle and a Coleman–Weinberg Realisation of the Localised Modulation in Late-Time Expansion History

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Abstract

This paper is the formal successor to v1.1 in the LVC working-paper series. The previous paper exhibited a half-twist sine–Gordon kink whose squared quarter-angle reproduces the observed sech^2 modulation at $\theta_c = \ln \varphi$ in the logarithmic-time coordinate $\theta = \ln(1 + z)$. The present paper does not extend the phenomenology; it records two distinct facts. First, the four structural requirements that select admissible modulation profiles in the LVC programme—localisation, compactness, asymptotic recovery, and low-parameter structure—can be stated independently of any specific Lagrangian, and the v1.1 sine–Gordon construction is one realisation of them rather than the unique one. Second, the z -coordinate derivative-of-Gaussian profile $T(z) = 1 + A(z/w) \exp[-(z/w)^2]$ documented in the v9 line of the phenomenological series admits an exact closed-form Lagrangian reconstruction. Under the field redefinition $\bar{\phi} = (A/2) \exp(-u^2)$ with $u = z/w$, the modulation is the static BPS-saturated solution of a canonical scalar with potential $V(\bar{\phi}) = 2\bar{\phi}^2 \ln(2\bar{\phi}/A)$. The potential is of the Coleman–Weinberg functional form and is generated at one loop by integrating out a coupled massless scalar with linear coupling $\frac{1}{2}\lambda\bar{\phi}\chi^2$; the $\bar{\phi}^2 \ln \bar{\phi}$ coefficient is universal at one loop, and the scale A enters only through mass renormalisation. On the PP-excluded data set ($N = 36$) the $w = 2$ lock construction returns $\chi^2 = 60.13$ at $k = 4$, against $\chi^2 = 78.96$ at $k = 3$ for Λ CDM, giving $\Delta\text{BIC} = -15.25$; free-fit w on the same data returns $w = 2.026$, locating $w = 2$ at the data minimum within 1.3%. The R6 realisation and the v1.1 sech^2 realisation share the same LVC structure but rest on distinct microscopic mechanisms—one classical and topological, the other quantum and effective—and produce sign-opposite shifts in the time-delay distance $D_{\Delta t}$ at two of the seven TDCOSMO lens systems. The reconstruction is offered as an exact existence result for an alternative realisation; $w = 2$ and the value of A are imposed inputs, not derived from the action.

1 Introduction

The phenomenological status of a localised modulation $T(z) = H(z)/H_{\Lambda\text{CDM}}(z)$ in the late-time expansion rate is documented across the v1–v16 working-paper series and the v0.1–v0.3 theoretical reconstruction line. The modulation is multiplicative, returns to unity at $z = 0$ and at high redshift, and exhibits a narrow excursion in a window centred near $z \approx 0.6\text{--}\sqrt{2}$ depending on the

parametrisation. v1.1 [1] recorded an exact topological reconstruction of the θ -coordinate sech^2 profile as a half-twist sine–Gordon kink.

The present paper does not extend the phenomenology and does not propose a fundamental theory. It records two facts and one structural observation. First, the four requirements that informally guided the v1.1 reconstruction—localisation, compactness, asymptotic recovery, and low-parameter structure—can be stated independently of the choice of action, and admit more than one realisation. Second, the z -coordinate derivative-of-Gaussian profile $T(z) = 1 + A(z/w) \exp[-(z/w)^2]$ catalogued as R6 in the phenomenological series admits a Lagrangian reconstruction at the same closure level as v1.1, but with a logarithmic Coleman–Weinberg potential rather than a sine–Gordon one. Third, the two realisations make sign-opposite predictions for the lensing time-delay distance $D_{\Delta t}$ in two of the seven TDCOSMO lenses.

The paper is organised in three parts. Part I (Section 2) records the four LVC requirements in a model-independent form. Part II (Sections 3–7) records the R6 Lagrangian reconstruction. Part III (Section 8) records the relation between the v1.1 and R6 realisations. Limitations are stated in Section 9.

The discipline of the paper is the same as v1.1: phenomenology and reconstruction are kept separate, and the reconstruction is an inverse program. Given a profile fixed empirically, an action whose static sector reproduces that profile is exhibited. No claim is attached to the choice of action beyond the consistency of its static solution with the previously fitted modulation.

Part I — The LVC organising principle

2 Four structural requirements

The LVC programme has selected, across the working-paper series, modulation candidates that meet four structural requirements. The requirements are stated below in a form independent of any specific Lagrangian. They were applied informally in v1.1; their explicit statement here is the only new content of Part I.

(L) Localisation. The modulation $T(z) - 1$ is concentrated in a finite redshift window and decays at both ends of the observed range:

$$T(z) - 1 \rightarrow 0 \quad \text{as } z \rightarrow 0 \text{ and as } z \rightarrow \infty, \tag{1}$$

with decay sufficiently rapid that no power-law tail extends the modulation beyond the window where the BAO data resolve it.

(C) Compactness. The fundamental scalar field of the Lagrangian reconstruction takes values in a compact range. The two realisations exhibited in this series are:

- v1.1: $\vartheta \in [0, 4\pi]$, a half-twist compact phase;
- R6 (this paper): $\bar{\phi} \in [0, A/2]$, a bounded scalar interval.

(A) Asymptotic recovery. $T(z) \rightarrow 1$ at $z = 0$ and at high redshift, recovering Λ CDM at both ends of the observable range. The modulation is an intermediate- z transient and not a global modification.

(P) Low-parameter structure. The full-lock construction is preferred to carry at most one imposed lock beyond H_0 and Ω_m . v1.1 has zero imposed locks in the modulation sector (its θ_c , A , w are all derived from $\ln \varphi$ and an internal relation); the R6 realisation has one imposed lock ($w = 2$).

The four requirements are not a derivation, and no claim of uniqueness is attached. They record the criteria under which both the v1.1 and R6 reconstructions have been retained as candidates, and against which any further reconstruction in this series is to be compared.

Part II — The R6 realisation

3 The R6 ansatz

The R6 modulation, catalogued in the v9 line of the phenomenological series, is

$$T(z) = 1 + A \cdot \frac{z}{w} \cdot \exp \left[- \left(\frac{z}{w} \right)^2 \right], \quad w = 2, \quad (2)$$

with A an amplitude parameter and w a width fixed at $w = 2$ in the lock construction. Writing $u \equiv z/w$ and $v(u) \equiv T(u) - 1$, the modulation deviation is

$$v(u) = A u e^{-u^2}, \quad (3)$$

which vanishes at $u = 0$ and at $u \rightarrow \infty$, reaches a maximum $v_{\max} = A/\sqrt{2e}$ at $u_{\text{peak}} = 1/\sqrt{2}$, and is non-monotonic on its domain.

4 Canonical field recovery

Direct attempts to interpret v itself as the fundamental scalar of a canonical Lagrangian fail because $v(u)$ is non-monotonic and the induced effective potential $V'_{\text{eff}}(v)$ becomes multi-valued. A field with single-valued potential is recovered by the following two-step redefinition.

Step 1: cumulative-phase field. Define

$$\psi(u) \equiv \int_0^u v(u') du' = \frac{A}{2} (1 - e^{-u^2}). \quad (4)$$

Since $v(u) > 0$ for all $u > 0$, ψ is monotonic on $[0, \infty)$ with range $[0, A/2]$.

Step 2: bounded scalar. Define

$$\bar{\phi}(u) \equiv \frac{A}{2} - \psi(u) = \frac{A}{2} e^{-u^2}. \quad (5)$$

The redefined field $\bar{\phi}$ is a Gaussian profile on $[0, \infty)$ in u with range $[0, A/2]$ in field space. The R6 modulation is recovered as the negative derivative,

$$v(u) = -\frac{d\bar{\phi}}{du} = A u e^{-u^2}. \quad (6)$$

5 Logarithmic potential

The action is

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{2\kappa} + \mathcal{L}_m + \mathcal{L}_{\bar{\phi}} \right], \quad \mathcal{L}_{\bar{\phi}} = \frac{1}{2} g^{\mu\nu} \partial_\mu \bar{\phi} \partial_\nu \bar{\phi} - V(\bar{\phi}), \quad (7)$$

with potential

$$\boxed{V(\bar{\phi}) = 2\bar{\phi}^2 \ln\left(\frac{2\bar{\phi}}{A}\right)}. \quad (8)$$

The Euler–Lagrange equation in the homogeneous static sector reads

$$\bar{\phi}'' + V'(\bar{\phi}) = 0, \quad V'(\bar{\phi}) = 4\bar{\phi} \ln(2\bar{\phi}/A) + 2\bar{\phi}, \quad (9)$$

with prime denoting d/du . Substituting $\bar{\phi}_{\text{static}}(u) = (A/2)e^{-u^2}$ verifies the equation as a symbolic identity; the numerical residual at representative u values is at the level of machine precision ($\sim 10^{-17}$). The first integral

$$\frac{1}{2}(\bar{\phi}')^2 + V(\bar{\phi}) = E_{\text{static}} \quad (10)$$

takes the value $E_{\text{static}} = 0$ everywhere on the static trajectory, the BPS-saturated condition. The boundary values $\bar{\phi}(0) = A/2$ and $\bar{\phi}(\infty) = 0$ coincide with the two zeros of $V(\bar{\phi})$ on the field range.

The potential has the following structural features.

- Zeros: $V(0) = 0$ (by $\bar{\phi}^2 \ln \bar{\phi} \rightarrow 0$) and $V(A/2) = 0$ (by $\ln(1) = 0$).
- Sign: $V(\bar{\phi}) < 0$ on the interior $0 < \bar{\phi} < A/2$.
- Critical point: $V'(\bar{\phi}_*) = 0$ at $\bar{\phi}_* = A/(2\sqrt{e})$, with $V(\bar{\phi}_*) = -A^2/(4e)$. This point coincides with the location of v_{max} on the static trajectory.
- Universal coefficient: Writing $V(\bar{\phi}) = 2\bar{\phi}^2 \ln \bar{\phi} + 2 \ln(2/A) \cdot \bar{\phi}^2$, the coefficient of the $\bar{\phi}^2 \ln \bar{\phi}$ term is 2 independently of A . The remaining $\bar{\phi}^2$ piece is a mass term whose coefficient depends on A only through $\ln(2/A)$.

The BPS integral

$$\int_0^{A/2} \sqrt{-2V(\bar{\phi})} d\bar{\phi} = \frac{\sqrt{\pi}}{8\sqrt{2}} A^2 \quad (11)$$

is independent of w . This is a structural difference from the v1.1 reconstruction, in which the BPS integral over the cubic effective potential $V_{\text{eff}}(v) = -(2/w^2)v^2(1 - v/A)$ depends on both w and A and supplies the derived relation $w = A/\ln \varphi$. In the R6 reconstruction no such derived relation between w and A is available: w enters only through the coordinate identification $u = z/w$ between the field-theory evolution parameter and the cosmological redshift.

6 One-loop interpretation

The functional form $\bar{\phi}^2 \ln \bar{\phi}$ in equation (8) matches the one-loop effective potential generated, in the Coleman–Weinberg framework, by a coupled scalar of mass $M^2(\bar{\phi}) = \lambda \bar{\phi}$. Specifically, augmenting the action by

$$\mathcal{L}_\chi = \frac{1}{2}(\partial\chi)^2 - \frac{1}{2}\lambda\bar{\phi}\chi^2 \quad (12)$$

and integrating out χ at one loop in $\overline{\text{MS}}$ yields

$$V_{\text{eff}}(\bar{\phi}) = V_{\text{tree}}(\bar{\phi}) + \frac{(\lambda\bar{\phi})^2}{64\pi^2} \left[\ln\left(\frac{\lambda\bar{\phi}}{\mu^2}\right) - \frac{3}{2} \right], \quad (13)$$

expanding to

$$V_{\text{eff}}(\bar{\phi}) = V_{\text{tree}}(\bar{\phi}) + \frac{\lambda^2}{64\pi^2} \bar{\phi}^2 \ln \bar{\phi} + (\text{scheme-dependent } \bar{\phi}^2 \text{ terms}). \quad (14)$$

Matching the $\bar{\phi}^2 \ln \bar{\phi}$ coefficient to that of (8) gives $\lambda^2/(64\pi^2) = 2$, i.e. $\lambda = 8\pi\sqrt{2}$. The $\bar{\phi}^2 \ln \bar{\phi}$ coefficient is scheme-independent at one loop; the accompanying $\bar{\phi}^2$ term is renormalisation-scheme-dependent and is absorbed into the tree-level mass parameter of V_{tree} .

Three observations qualify the matching.

- (i) The matching identifies the functional form, not a fully derived UV theory. The required χ field is massless, and the V_{tree} that absorbs the scheme-dependent $\bar{\phi}^2$ counterterm is not specified.
- (ii) The coupling value $\lambda = 8\pi\sqrt{2} \approx 35.5$ is large; resummation of higher-loop corrections is not addressed here.
- (iii) Standard Coleman–Weinberg in ϕ^4 theory produces $\phi^4 \ln \phi$ rather than $\phi^2 \ln \phi$. The functional difference traces to the linear mass coupling $M^2(\bar{\phi}) = \lambda\bar{\phi}$ rather than $\lambda\bar{\phi}^2$.

The interpretation in equation (13) is offered as the natural quantum-field-theoretic origin of the potential’s functional form. It is not used elsewhere in the paper.

7 Observational behaviour

The fit is performed on the PP-excluded data set: DESI DR2 BAO [2], DESI DR1 BAO [2], BOSS DR12 [3], eBOSS DR16 [4], Planck 2018 distance priors [5], and SH0ES H_0 [6], totalling $N = 36$ data points. The same fitting pipeline as v1.1 is used; the r_d scaling is fixed by $\theta_* \cdot D_M(z_*)$ from the recombination redshift z_* in each model. Pantheon+ supernovae are not included in the present fit; PP-included results are deferred.

Table 1 records the fits.

Three observations follow from Table 1.

Observation 1: $w = 2$ is at the data minimum. The free-fit width on the same data set returns $w_{\text{best}} = 2.026$, 1.3% from the imposed $w = 2$ lock. The χ^2 change between $w = 2$ locked and w free is $\Delta\chi^2 = 0.001$, indicating that the lock is at the data preference within the resolving power of the present datasets.

Model	k	χ^2	ΔBIC	Notes
ΛCDM	3	78.962	0 (baseline)	$H_0 = 70.34$, $\Omega_m = 0.277$
sech^2 full lock	2	56.134	-26.41	v1.1 reproduction
R6 ($w = 2$ lock)	4	60.128	-15.25	$A = 0.056$ free
R6 (w free)	5	60.127	-11.67	$w_{\text{best}} = 2.026$

Table 1: PP-excluded ($N = 36$) fits. $\Delta\text{BIC} \equiv \text{BIC} - \text{BIC}_{\Lambda\text{CDM}}$.

Observation 2: R6 BIC evidence is weaker than v1.1. The R6 lock construction returns $\Delta\text{BIC} = -15.25$ on the same data set on which v1.1 returns $\Delta\text{BIC} = -26.41$. Both values constitute strong Bayesian evidence over ΛCDM by standard thresholds (decisive evidence is conventionally $|\Delta\text{BIC}| > 10$). The difference of 11.16 between the two reconstructions traces to a $\Delta\chi^2 = 3.99$ and a difference of two locked parameters.

Observation 3: PP-included extrapolation. The v1.1 PP-included result on $N = 1738$ returns $\Delta\text{BIC} = -30.31$, a strengthening of -3.90 over the PP-excluded value. A proportional PP-included strengthening for R6 is plausible but not established by the present analysis. Section 9 records this as a deferred item.

Lens time-delay distance. The same R6 modulation produces a measurable shift in the strong-lensing time-delay distance

$$D_{\Delta t} = (1 + z_l) \frac{D_A(z_l) D_A(z_s)}{D_A(z_l, z_s)} \quad (15)$$

across the seven lenses of the TDCOSMO sample [8]. Table 2 records the pure-modulation effect on $D_{\Delta t}$ computed at fixed base parameters $H_0 = 70$, $\Omega_m = 0.30$ for both reconstructions, relative to ΛCDM at the same base parameters. The pure-modulation calculation is the appropriate comparison for the model identification question; fits with the model-specific best-fit H_0 , Ω_m are dominated by the difference in those best-fit values and are not reported here.

Lens	z_l	z_s	$\text{sech}^2 \Delta D_{\Delta t} / D_{\Delta t}^{\Lambda\text{CDM}}$	R6 $\Delta D_{\Delta t} / D_{\Delta t}^{\Lambda\text{CDM}}$
RXJ1131-1231	0.295	0.654	+1.268%	+0.018%
PG1115+080	0.311	1.722	+0.288%	-0.056%
B1608+656	0.630	1.394	-1.247%	-0.103%
HE0435-1223	0.456	1.693	+0.036%	-0.099%
SDSSJ1206+4332	0.745	1.789	-2.332%	-0.261%
WFI2033-4723	0.658	1.662	-1.670%	-0.178%
DES J0408-5354	0.597	2.375	-1.224%	-0.322%
mean			-0.697%	-0.143%

Table 2: Time-delay distance shift relative to ΛCDM , computed at fixed base parameters $H_0 = 70$, $\Omega_m = 0.30$. The two reconstructions use the lock values $A_{\text{sech}^2} = (\ln \varphi)^4$, $\theta_c = \ln \varphi$, $w_{\text{sech}^2} = (\ln \varphi)^3$ for v1.1 and $A_{\text{R6}} = 0.0766$, $w_{\text{R6}} = 2$ for R6.

Two facts from Table 2 are recorded.

Observation 4: R6 mean shift is smaller than sech^2 . The mean shift across the seven lenses is -0.143% for R6 and -0.697% for sech^2 . The ratio is approximately 5; the R6 maximum effect (-0.322% at DES J0408) is smaller than the sech^2 minimum effect on the same lens.

Observation 5: sign-opposite predictions on two lenses. At PG1115+080 and at HE0435–1223, the R6 and sech^2 predictions for $\Delta D_{\Delta t}/D_{\Delta t}^{\Lambda\text{CDM}}$ have opposite signs ($+0.288\%$ vs. -0.056% , and $+0.036\%$ vs. -0.099% , respectively). Both lenses share the structural feature $z_l \in [0.3, 0.5]$, $z_s \approx 1.7$. The two reconstructions place their modulation peak at $z = 1/\varphi \approx 0.62$ and $z = \sqrt{2} \approx 1.41$ respectively; the lens-redshift and source-redshift ranges of the TDCOSMO sample sample the two peaks differently, and the ratio of effects on $D_A(z_l)$ versus $D_A(z_s)$ inverts on the two lenses listed.

The magnitudes in Table 2 are at the percent level; the TDCOSMO seven-lens combined $D_{\Delta t}$ precision is currently at the three-percent level [8]. The discrimination implied by Observation 5 is therefore not at present statistically resolved, but is recorded as a deferred test.

Part III — Relation to v1.1

8 Two realisations of the same organising principle

The R6 reconstruction (Part II) and the v1.1 sine–Gordon reconstruction are two realisations of the four LVC requirements stated in Part I. The following table records the parallel structure.

	v1.1 (Branch A)	R6 (Branch B, this paper)
Coordinate	$\theta = \ln(1 + z)$	$u = z/w$
Profile family	sech^2 on θ	derivative-of-Gaussian on z
Field	ϑ (compact phase)	$\bar{\phi}$ (bounded scalar)
Field range (C)	$\vartheta \in [0, 4\pi]$	$\bar{\phi} \in [0, A/2]$
Lagrangian	$\frac{1}{2}(\partial\vartheta)^2 - m^2[1 - \cos(\vartheta/2)]$	$\frac{1}{2}(\partial\bar{\phi})^2 - 2\bar{\phi}^2 \ln(2\bar{\phi}/A)$
Potential class	trigonometric	logarithmic (Coleman–Weinberg form)
Static-sector mechanism	topological soliton (kink)	BPS-saturated bounce
Loop order of origin	tree level	one loop
Modulation as image	$v = \varepsilon^2 \sin^2(\vartheta_{\text{kink}}/4)$	$v = -\partial_u \bar{\phi}$
Imposed locks beyond H_0, Ω_m	0 (θ_c, A, w all derived)	1 ($w = 2$)
Peak location in z	$z \approx 1/\varphi \approx 0.618$	$z = \sqrt{2} \approx 1.414$
PP-excluded ΔBIC ($N = 36$)	-26.41	-15.25
TDCOSMO mean $\Delta D_{\Delta t}$	-0.697%	-0.143%
Sign-opposite lenses	—	2 of 7 (PG1115, HE0435)

Table 3: The two realisations of the LVC requirements documented in the present series.

Three facts from the comparison are recorded.

Same organising principle. Both reconstructions satisfy the four requirements of Section 2. The selection of profile family within each branch (sech^2 in branch A, derivative-of-Gaussian in branch B) is not derived from the requirements; it is fixed empirically in the phenomenological series.

Different microscopic mechanism. The v1.1 mechanism is classical and topological: the static configuration is a soliton with a conserved 4π winding number, and stability is topological. The R6 mechanism is quantum-effective: the potential arises at one loop, and the static configuration is a BPS-saturated bounce with $E_{\text{static}} = 0$. The two mechanisms do not interpolate: a sech^2 profile cannot be written as $-\partial_u \bar{\phi}$ for any $\bar{\phi}$ admitting the logarithmic potential, and a derivative-of-Gaussian profile cannot be written as $\sin^2(\vartheta/4)$ for any sine-Gordon kink.

Sign-opposite realisation at the observable level. The two reconstructions are not distinguished by the BAO data set used for the fits in Table 1: both return strong Bayesian evidence over ΛCDM , with v1.1 supplying -11.16 additional BIC support traceable to one additional locked parameter and one improved χ^2 . The two are distinguished by the time-delay distance: the predicted shifts have opposite signs on two of the seven TDCOSMO lenses (Observation 5). The discrimination is below current TDCOSMO precision; whether it is resolved by future precision lensing is recorded as a deferred test.

9 Limitations and deferred items

The R6 reconstruction is offered as an existence result for an alternative LVC realisation. Five items are recorded as limitations or deferred.

Imposed inputs. The width $w = 2$ and the amplitude A are imposed inputs. No internal principle of the action selects $w = 2$ among nearby integer or natural values, and no derived relation between w and A is available in the R6 construction. The selection principle that would fix them, as in v1.1, remains open. This places R6 one imposed lock behind v1.1, which fixes θ_c, A, w all internally.

PP-included fit deferred. The fit in Table 1 is on the PP-excluded data set ($N = 36$). The corresponding PP-included fit ($N = 1738$, including Pantheon+ unbinned supernovae) is not performed in this paper. v1.1 documents a PP-included strengthening of -3.90 in ΔBIC over its PP-excluded value; whether R6 strengthens proportionally is not established here.

Hubble friction. The Lagrangian static sector $\bar{\phi}'' + V'(\bar{\phi}) = 0$ neglects the Hubble friction term $3H\dot{\bar{\phi}}$ that enters the FLRW equation of motion. v1.1 documents the size of the percent-level correction in the sech^2 case (1.98% at one width); the corresponding correction for R6 is not computed.

Explicit one-loop derivation. The matching in Section 6 quotes the Coleman–Weinberg one-loop formula. An explicit derivation in the specific UV theory $\mathcal{L}_{\text{UV}} = \mathcal{L}_{\bar{\phi}} + \frac{1}{2}(\partial\chi)^2 - \frac{1}{2}\lambda\bar{\phi}\chi^2$, with

all renormalisation counterterms specified, is not given in this paper. The large value $\lambda = 8\pi\sqrt{2}$ raises a separate question about the validity of the one-loop approximation, not addressed here.

Lens-specific posteriors. The TDCOSMO shifts in Table 2 are pure-modulation effects at fixed base parameters. A full posterior analysis with lens-specific likelihoods [8] is not performed; the seven-lens combined $D_{\Delta t}$ posterior for R6 is deferred.

10 Outlook

The two facts recorded in this paper—that the four LVC requirements admit more than one realisation, and that the R6 derivative-of-Gaussian profile closes at the Lagrangian level with a Coleman–Weinberg potential—do not select between the v1.1 and R6 reconstructions. Both remain open candidates for the underlying microscopic origin of the modulation. The discriminating observable identified is the lensing time-delay distance at the percent level; whether this discrimination is resolved by future precision lensing is the empirical question left open by the present paper.

The status of the LVC programme after the present paper is therefore as follows: the organising principle admits multiple realisations, the v1.1 and R6 reconstructions exhaust the two retained in this series, and the two are observationally complementary at sub-percent strong-lensing precision. Further realisations satisfying the four requirements are not excluded.

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